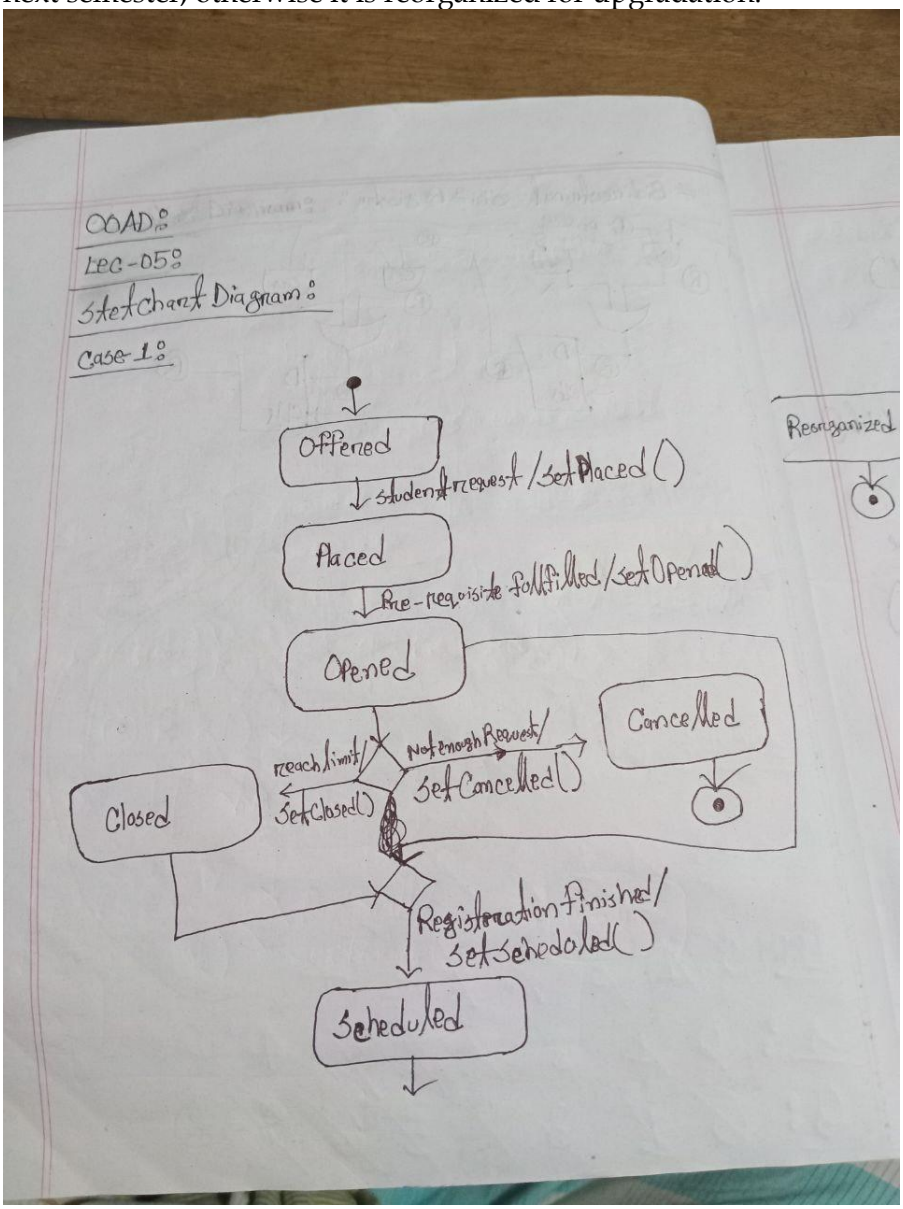
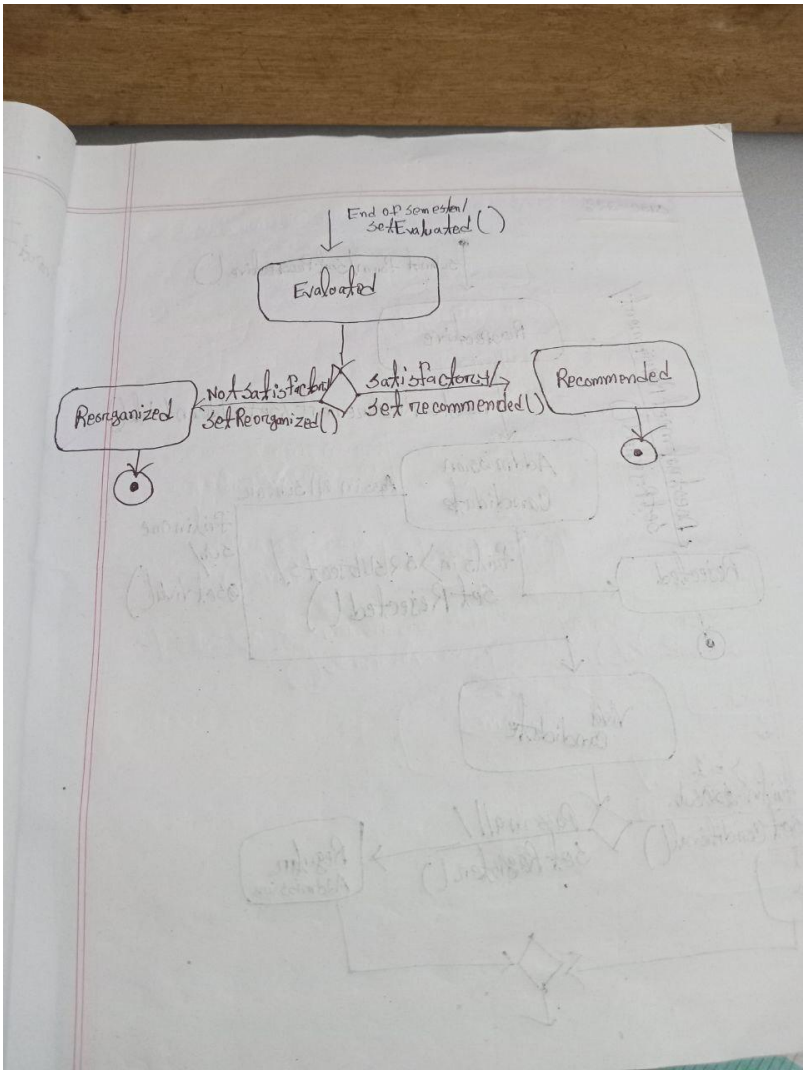
Statechart Diagram

Case 1

A course is initially created in the offered state. Students request the courses they want to register, and those courses are placed for registration. If the pre-requisites are fulfilled for those courses, they are opened for enrolling. When a course reaches the limit of its capacity the course is then closed for further student requests. On the other hand, if there are not enough students requesting a course the course is cancelled. Once the student registration is finished, all the opened and closed courses are scheduled. At the end of the semester the course is evaluated by the authority. If the evaluation is satisfactory it is recommended for the next semester, otherwise it is reorganized for upgradation.

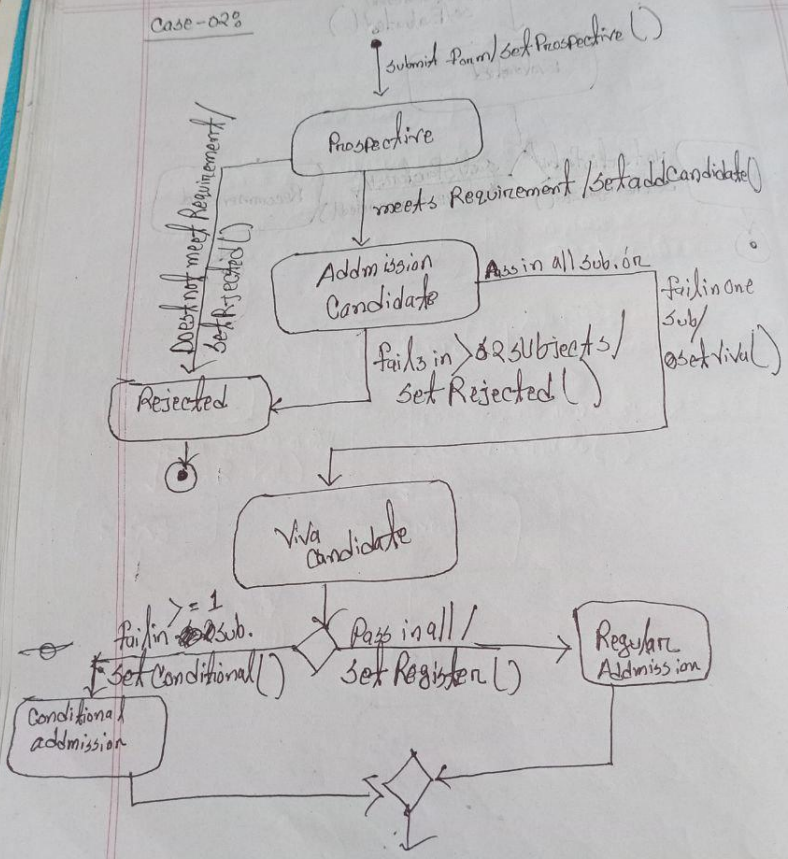


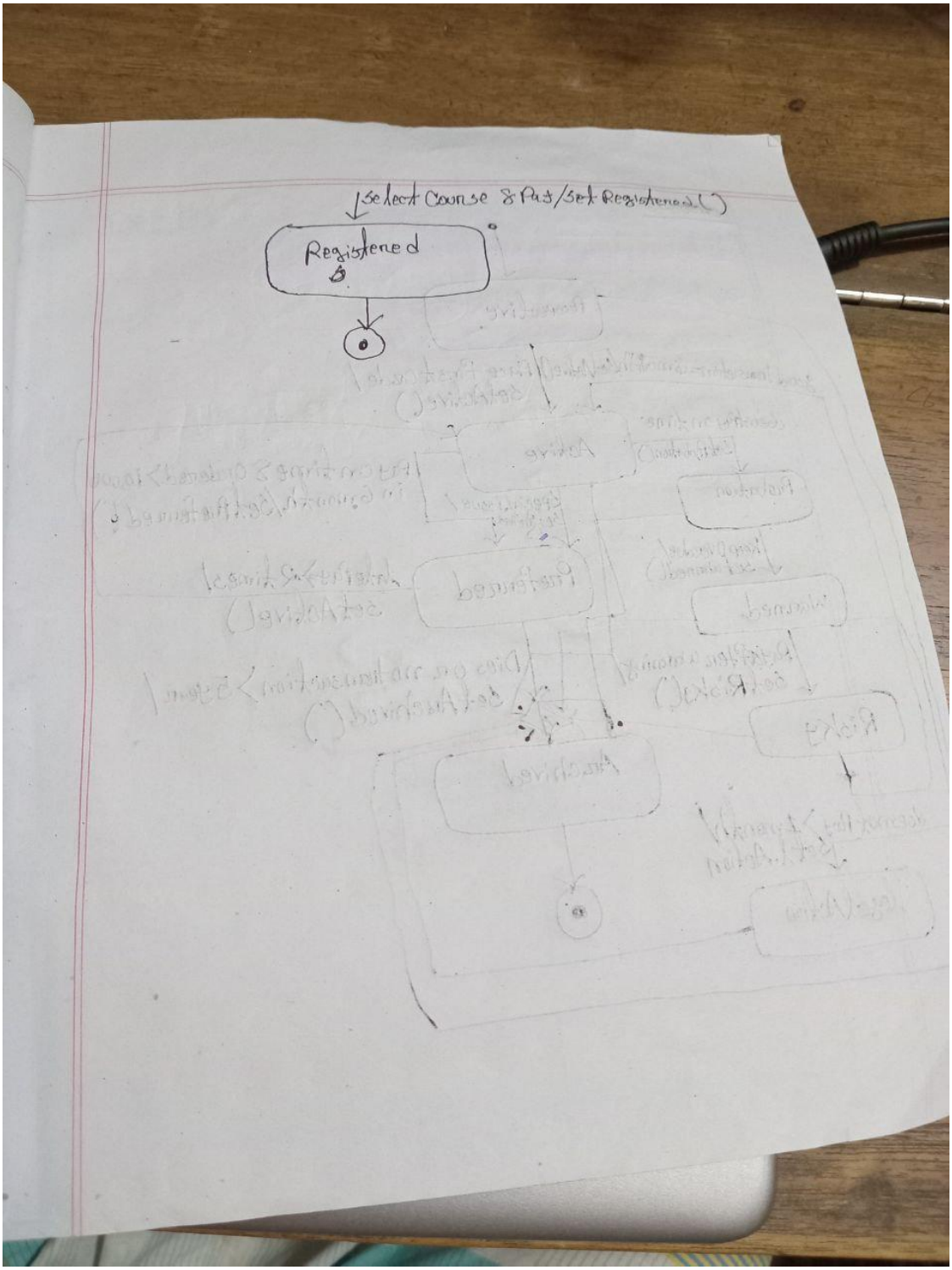


Case 2

A student object is created as prospective student when he submits the admission form to a university. Student information is verified and if he meets all the requirements, he is asked to sit for an admission test as an admission candidate. But if the student doesn't meet the requirements, he is rejected, and the student object is deleted permanently from the system. Any student who fails in more than two subjects in the admission exam is also rejected. Students who pass in all the subjects or fail in one or two subjects are called to attend viva as viva candidate. At the viva the students who failed in one or more subjects are given conditional admission with non-credit courses. Students who pass all the subjects are given regular admission. All sorts of admitted students are required to go through pre-advising to select appropriate courses for the semester. Once courses are selected, students pay for the courses they have taken and become registered students of the semester.

Case-02:

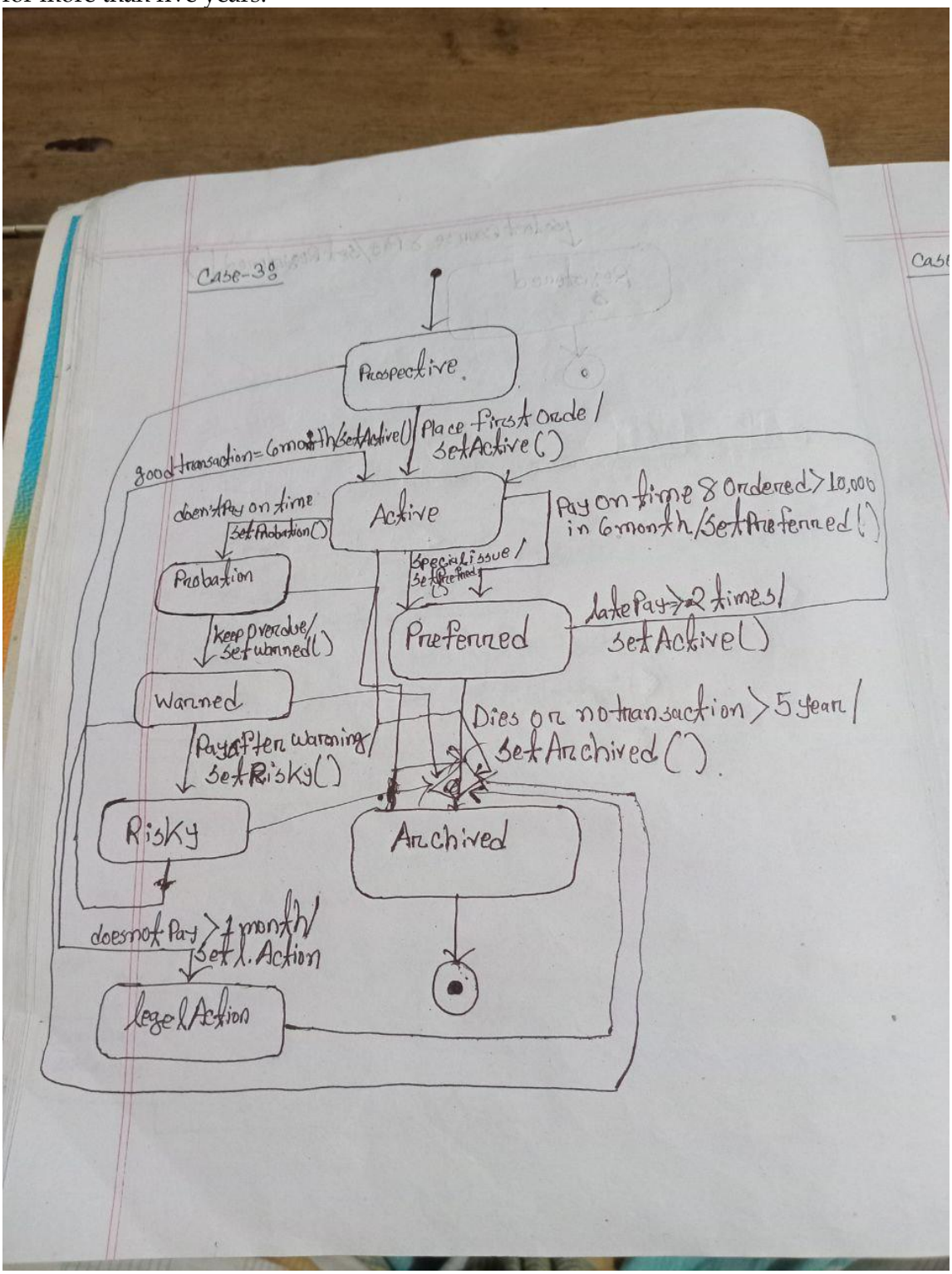

submit Form/ Set Prospective ()
Prospective
meets Requirement /SetaddCandidate()
Doesn't not meet Requirement/ SetRejected()
Addmission Candidate
Ass in all sub. or
fail in one sub/ setViva()
fails in >2 subjects/ Set Rejected()
Rejected
Viva candidate
fail in >=1 sub.
Set Conditional()
Pass in all/ Set Register()
Regular Addmission
Conditional addmission



Case 3

In an order processing system, all customers are initially set up as prospective customers, but when they place their first order, they are considered active. If a customer doesn't pay an invoice on time, he is placed on probation. If he continues to keep overdue, he is warned. If the customer doesn't pay the dues within a month of the warning, legal actions are taken against the customer. If the customer pays after the warning, he is taken back to the system as a risky customer. Only when he has a good transaction record for six months, he becomes an active customer. If an active customer pays on time and has ordered more than $10,000 in the previous six months, he warrants preferred status. Preferred status may be changed only if the customer is late on two or more payments. Then he returns to active status rather than probation, giving him the benefit of the doubt based on his preferred history. Sometimes an active customer can become a preferred customer if the management has special issues with the customer. The customer

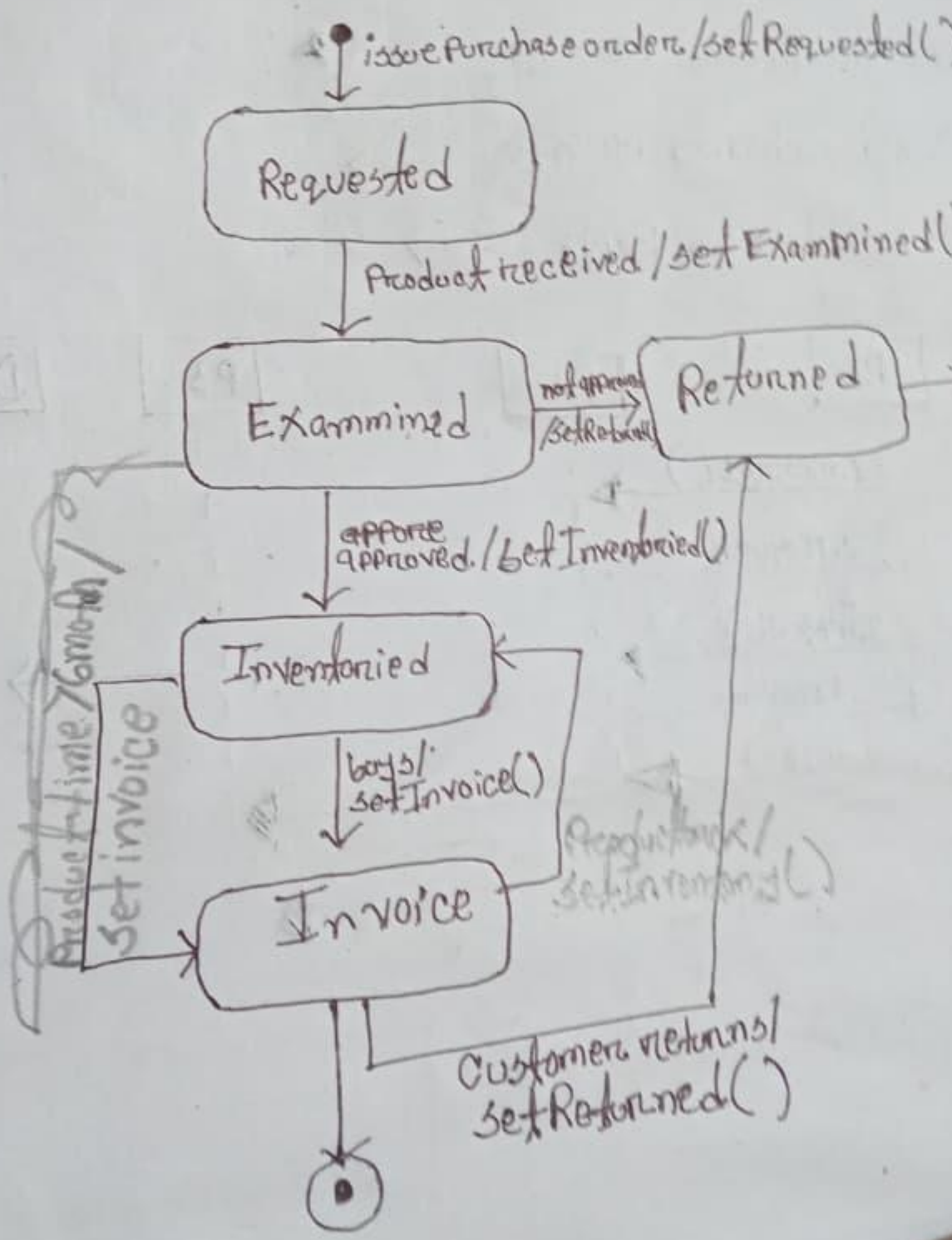
information is archived for good if the customer dies or if there has been no transaction with the customer for more than five years.



Case 4

Products are first entered into our system as requested when a purchase order for that product is issued. The purchase order is then printed and recorded with that product. The purchase order is then sent to the vendor. When the product is received from the vendor, it is examined by the quality control department. Once the product is approved by the quality control department, it is placed into inventory by recording the location where it is placed. If it is not approved the product is returned to the vendor. When a customer buys a product, it is included in an invoice. Sometimes, a product is returned. In that case, you put the product back into inventory and record the location. If a product is lying in the inventory for more than six months it is returned to the vendor, but this time the vendor buys back the product in a reduced price, which means the product is included in an invoice.

Case-4:


issue Purchase order. /set Requested()
Requested
Product received / set Examined()
Examined
not approved /SetReturned()
Returned
approved. / Set Inventoried()
Inventoried
buys/ set Invoice()
Invoice
Set invoice
Customer returns/ setReturned()